

STRUCTURED QUESTIONS (100 MARKS)

(INSTRUCTION: This question booklet has 7 questions. Please write the answers for all questions in the answer booklet. Show all your working steps.)

QUESTION 1 [15 Marks]

- (a) Consider the initial contents for some registers as given in **Figure 1**. Assume that the two lines of instructions given in **Figure 2** are executed based on these initial contents. Calculate the final contents of registers EAX, EBX, and EDX after the instructions have been executed. [4 Marks]

EAX=770C0810	EBX=7FFD0410	ECX=00000000	EDX=00400001
ESI=00000000	EDI=00000000	EBP=0012FF94	ESP=0012FF8C
EIP=00401025	EFL=00000A82	CF=0	SF=1 ZF=0 OF=1

EAX: 770C0200

Figure 1
Initial registers

0200×0410
 $= 00082000$
 DX: AX
 * does not change

MOV AX, 200h
 MUL BX

4 bit replace with

EAX = 770C2000
 EBX = 7FFD0410
 EDX = 00400008

Figure 2

- (b) Consider a 16 bits system that uses even parity. Calculate $8028h + 803Ah$ in binary and based on the result, determine the value for each of the flags shown in **Table 1**. [4 Marks]

Table 1

ZF	SF	CF	OF	AF	PF
0	0	1	1	1	0

$8028h = 1000\ 0000\ 0010\ 1000$
 $803Ah = 1000\ 0000\ 0011\ 1010$

$$\begin{array}{r}
 1000\ 0000\ 0010\ 1000 \\
 +\ 1000\ 0000\ 0011\ 1010 \\
 \hline
 0000\ 0000\ 0110\ 0010
 \end{array}$$

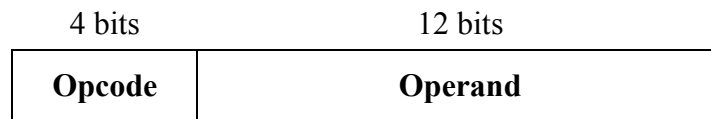
low nibble to high nibble = AF = 1
 (-) (-) (+)

3 1's → odd parity = 0
 positive sign = 0
 carry

pf → not include the overflow

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(c) Given an ISA (Instruction Set Architecture) format is as in **Figure 3**:



Opcode: $0001_2 = \text{MOV}$, $0101_2 = \text{SUB}$

Figure 3

Trace the execution of instruction cycle for memory address 39Dh as shown in **Table 2**. Given the initial values as in **Table 3**, complete the instruction cycle by writing the contents of the related CPU registers with the respective microoperations. [7 Marks]

Table 2

Memory Address	Memory Content (Hex)	Instruction/Data
39C	1743	MOV AX, 9Ah
39D	5743	SUB AX, NUM
39E		
...	...	...
743	000A	NUM
744	0F00	

value in AX
Subtract with
value NUM and store
in AX

Table 3

Clock	IP/PC	MAR	MBR	IR	AX	Microoperations
t_0	39D				009A	IP/PC = 39D
t_1	39E	39D			009A	MAR ← IP/PC IP/PC ← IP/PC + 1 MBR ← [MAR]
t_2	39E	39D	5743		009A	IR ← MBR
t_3	39E	39D	5743	5743	009A	IR ← MBR
t_4	39E	743	000A	5743	009A	MAR ← 743 MBR ← [MAR]
t_5	39E	743	000A	5743	0090	SUB: AX ← AX - MBR
t_6	39F	39E	000A	5743	0090	MAR ← IP/PC IP/PC ← IP/PC + 1

need operand 743
is load in MAR

Start decode
opcode → 5
operand → 743

end
here!!

QUESTION 2 [17 Marks]

- (a) **Table 4** illustrates the segments of a pipeline with the registers' interface delay is 5 ns. Based on the given information in **Table 4**, calculate the following:

Table 4

Segment	Execution Times (ns)
Fetch Instruction (FI)	50
Decode Instruction (DI)	45
Fetch Operand (FO)	40
Execute Instruction (EI)	45
Write Operand (WO)	35

- The cycle time of non-pipeline and pipeline. [4 Marks]
 - The execution time of 100 tasks for non-pipeline and pipeline. [4 Marks]
 - The real Speedup, S . [2 Marks]
 - The maximum Speedup, $S_{\max}$. [1 Mark]
- (b) Consider the assembly program in **Figure 4** is executed in a pipeline that consists of these segments: *Fetch Instruction* (FI), *Decode Instruction* (DI), *Execute Instruction* (EI) and *Write Operand* (WO). Based on the instructions given in **Figure 4**, determine which instruction(s) have the following hazards and rewrite the instructions using the proposed solutions.

I1:	MOV	AX, BX
I2:	ADD	CX, AX
I3:	INC	DX
I4:	JMP	TARGET
I5:	MOV	BX, 5
I6:	INC	CX
.		
I12: TARGET	DEC	AX

Figure 4

- Data dependency and solve using Delayed Load. [3 Marks]
- Branch difficulty and solve using Delayed Branch and Rearranging Instructions. [3 Marks]

$$t_p(12 + (n-1))$$

(a) Table 4 illustrates the segments of a pipeline with the registers' interface delay is 5 ns. Based on the given information in Table 4, calculate the following:

Table 4

Segment	Execution Times (ns)
Fetch Instruction (FI)	50
Decode Instruction (DI)	45
Fetch Operand (FO)	40
Execute Instruction (EI)	45
Write Operand (WO)	35

- The cycle time of non-pipeline and pipeline. [4 Marks]
- The execution time of 100 tasks for non-pipeline and pipeline. [4 Marks]
- The real Speedup, S . [2 Marks]
- The maximum Speedup, S_{max} . [1 Mark]

non pipeline
 $(50+45+20+45+35) \times 5$
 $t_n = 220 \text{ ns}$
 pipeline
 $50+5$
 $t_p = 55 \text{ ns}$
 $T_n = t_n \times k$
 $= 100 \times 220$
 $= 22000$
 $T_p = t_p (k + (n-1))$
 $= 55 (5 + (100-1))$
 $= 5720$
 $S = \frac{T_n}{T_p} = \frac{22000}{5720}$
 $= 3.846$

(b) Consider the assembly program in Figure 4 is executed in a pipeline that consists of these segments: Fetch Instruction (FI), Decode Instruction (DI), Execute Instruction (EI) and Write Operand (WO). Based on the instructions given in Figure 4, determine which instruction(s) have the following hazards and rewrite the instructions using the proposed solutions.

* Compare 2 instruction, kalau ada sama, kener add NOP

I1:	MOV	AX, BX
I2:	ADD	CX, AX
I3:	INC	DX
I4:	JMP	TARGET
I5:	MOV	BX, 5
I6:	INC	CX
...		
I12:	TARGET	DEC AX

data hazard

Figure 4

use NOP

- Data dependency and solve using Delayed Load. [3 Marks]
- Branch difficulty and solve using Delayed Branch and Rearranging Instructions. [3 Marks]

Just example

I1: MOV AX, BX
 NOP
 I2: ADD CX, AX

(i) Delayed branch
 $NOP = \text{no. of segment} - 1$
 $4 - 1$
 $NOP = 3$

Rearranging instruction

$$= 4 - 1$$

$$= 3$$

I4: JMP TARGET

NOP
 NOP
 NOP

I5: MOV BX, 5

JMP TARGET
 MOV AX, 1
 ADD CX, BX
 INC DX

iv) $S_{max} = k$
 $= 5$

	1	2	3	4	5		12
FI	I1	NOP	I2				
DI		I1	NOP	I2			
EI			I1	NOP	I2		
WO				I1	NOP	I2	

Example 18 : Solution for branching using rearranging instructions.

When the compiler detect a branch, it will rearrange the instructions so that there is no wasted pipeline.

How to arrange, moving the instruction up?

MOV
INC
ADD
RET I1

I1:

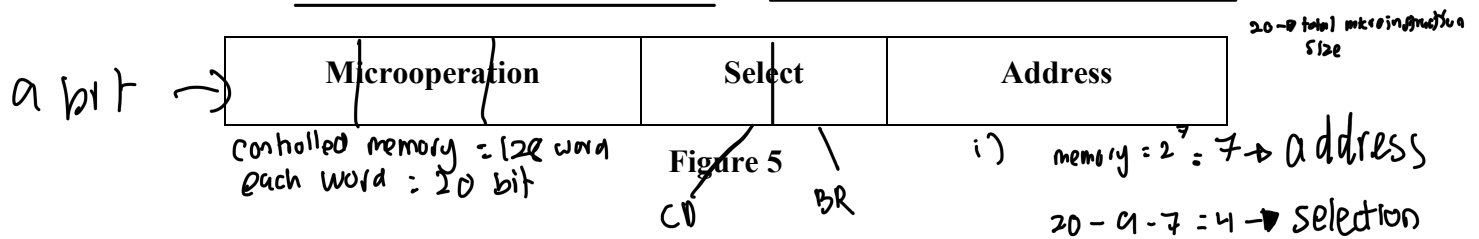
RET I1
MOV
INC
ADD

I1:

= No of segment - 1
 $= k - 1$
 $= 4 - 1 = 3$ (move up 3)

QUESTION 3 [12 Marks] *each microinstruction specifies single micro-operation to be performed*

Consider a system uses vertical microinstruction format with a control memory that consists of 128 words and each word consists of 20 bits as illustrated in **Figure 5**. The system also consists of 4 status conditions for branching and 9 bits microoperation with 3 fields.



(a) Based on **Figure 5**, calculate the size (in bits) of the following:

i) The Select and Address fields. [3 Marks]

ii) The Branch bits in the Select field. [3 Marks]

(b) Based on the answer in (a), and the symbolic microinstructions given in **Table 5**, complete the 20 bits microinstruction fields as listed in **Table 6**, for microinstructions (i), (ii) and (iii). [6 Marks]

Table 5

	Label	Address	Microinstruction			
(i)	ADD	0000100	NOP	I	CALL	INDRCT
(ii)			READ	U	JMP	<u>NEXT</u>
(iii)			ADD	U	JMP	<u>FETCH</u>
	FETCH	<u>1000000</u>	...			
	INDRCT	1000011	...			

(Remark: Use the binary code for microinstruction fields in **Appendix A** on page 11.)

Table 6

i) nop in
F1 = 000
F2 = 000
F3 = 000

	F1	F2	F3		
	Microoperation (F1, F2, F3)			Select (CD, BR)	Address
(i)	000	000	000	01 01	0000 (00)
(ii)	000	100	000	00 00	0000 101
(iii)	001	000	000	00 00	1000 000

1) CD: 01
BR: 01

this one given
cycle next

QUESTION 4 [12 Marks]

(a) Consider the following different types of memory:

- Hard Disk Drive (HDD)
- Dynamic Random Access Memory (DRAM)
- Static Random-Access Memory (SRAM)
- Registers

i) List the memory types based on memory pyramid hierarchy position from top to bottom. [2 Marks]

ii) Based on memory pyramid hierarchy, briefly explain how speed, and cost per bit differs from top to bottom. [2 Marks]

(b) Consider the main memory is divided into blocks. The memory word is 8 bits and the size of a block is 8 words.

i) What is the capacity of the main memory, if the total number of blocks in the memory is 128? [2 Marks]

ii) How many blocks in the main memory if the memory capacity is 64 Kbits? [2 Marks]

a) i) Register, SRAM, DRAM, HDD

ii) top to bottom
 speed: speed decrease, register fastest because they are directly inside CPU, HDD slowest

iii) cost decreased

(b) Consider the main memory is divided into blocks. The memory word is 8 bits and the size of a block is 8 words.

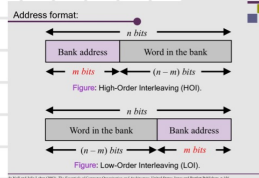
i) What is the capacity of the main memory, if the total number of blocks in the memory is 128? [2 Marks]

ii) How many blocks in the main memory if the memory capacity is 64 Kbits? [2 Marks]

$$\text{Block size} = 8 \text{ words} \times 8 \text{ bit} = 64 \text{ bits}$$

$$\begin{aligned} \text{memory capacity} &= \text{Block} \times \text{word} \times \text{bits} \\ &= 128 \times 8 \times 8 \\ &= 8192 \text{ bits} \end{aligned}$$

$$\begin{aligned} \text{ii) } 64 \text{ K} &= \text{block} \times 8 \times 8 \\ &= 1 \text{ K block} \\ &= 1024 \text{ block} \end{aligned}$$



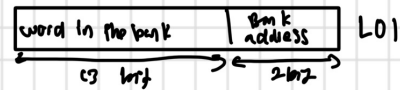
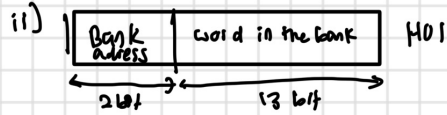
(c) Consider the memory capacity is 32 Kbytes and the total of banks is 4.

i) Calculate the bits for the Word in bank and the Bank address. [2 Marks]

ii) Based on the answer in (i), draw the address format used for both Low-Order Interleaving (LOI) and High-Order Interleaving (HOI). [2 Marks]

ii) $\text{memory capacity} = 32 \times 1024 = 32768 = 2^{15} \rightarrow 15 \text{ bit}$

$$\begin{aligned} 4 &\rightarrow 2^2 \rightarrow 2 \text{ bit bank address} \\ \text{Bank capacity} / \text{word in the bank} &= 15 - 2 \text{ bit} = 13 \text{ bit} \end{aligned}$$



- (c) Consider the memory capacity is 32 Kbytes and the total of banks is 4.
- Calculate the bits for the Word in bank and the Bank address. [2 Marks]
 - Based on the answer in (i), draw the address format used for both Low-Order Interleaving (LOI) and High-Order Interleaving (HOI). [2 Marks]

QUESTION 5 [15 Marks]

- (a) A main memory contains 32 words while the cache has only 8 words.
- Using Direct mapping, identify the fields of the main memory address. [1 Mark]
 - Table 7** illustrates a main memory and its content for some addresses. Complete **Table 8** as they respond to a series of request (in hexadecimal address) as in the following. [5 Marks]

19, 1A, 15, 12, 19

Table 7

Address (Hex)	Address (Binary)	Content (Hex)
1A	11010	B1
19	11001	A1
18	11000	F1
17	10 111	B2
16	10 110	E1
15	10 101	C1
14	10100	A2
13	10011	C2
12	10010	D1

Table 8

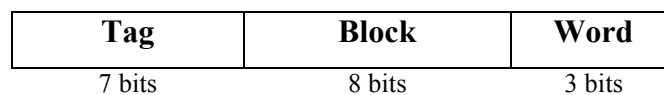
Line	Tag	Content (Hex)	Tag	Line	Hit	Miss	Update Cache	Read	Write
111									
110									
101									
100									
011									
010									
001									
000									

(i) Cache memory

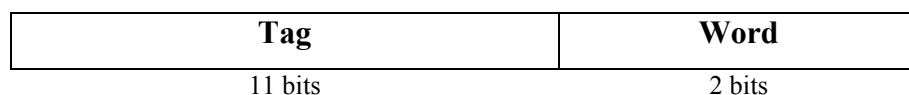
(ii) Request

(iii) Read/Write operation cache

- (b) Given the main memory address format using Block Direct mapping is shown in **Figure 6**. Consider each main memory word is 16 bits.

**Figure 6**

- Calculate the main memory capacity. [2 Marks]
 - Calculate the total cache words. [1 Mark]
 - Calculate the cache word size (in bits). [2 Marks]
- (c) Consider a main memory address format is given in **Figure 7**. Supposed the program generates the memory address 167Bh. Determine the tag and word in cache memory using Fully Associative mapping. [2 Marks]

**Figure 7**

(a) A main memory contains 32 words while the cache has only 8 words.

i) Using Direct mapping, identify the fields of the main memory address. [1 Mark]

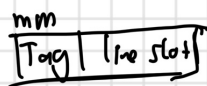
ii) Table 7 illustrates a main memory and its content for some addresses. Complete Table 8 as they respond to a series of request (in hexadecimal address) as in the following. [5 Marks]

19, 1A, 15, 12, 19

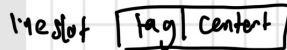
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17	10 111	B2
16	10 110	E1
15	10 101	C1
14	10100	A2
13	10011	C2
12	10010	D1

i) Direct mapping



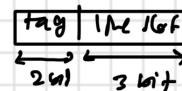
cache



$$32 = 2^5 = 5 \text{ bit}$$

$$8 = 2^3 \rightarrow 3 \text{ bit} \rightarrow \text{line slot}$$

$$5 - 3 = 2 \text{ bit} \rightarrow \text{tag}$$



ii)

Table 8

Line	Tag	Content (Hex)	Tag	Line	Hit	Miss	Update Cache	Read	Write
111			11	001		/	/	/	
110			11	010		/	/	/	
101	10	C1	10	011		/	/	/	
100			10	100		/	/	/	
011			11	001	/			/	
010	10	D1							
001	11	A1							
000									

(i) Cache memory

(ii) Request

(iii) Read/Write operation cache

(b) Given the main memory address format using Block Direct mapping is shown in Figure 6. Consider each main memory word is 16 bits.

Tag	Block	Word
7 bits	8 bits	3 bits

Figure 6

i) Calculate the main memory capacity. [2 Marks]

ii) Calculate the total cache words. [1 Mark]

iii) Calculate the cache word size (in bits). [2 Marks]

(c) Consider a main memory address format is given in Figure 7. Supposed the program generates the memory address 167Bh. Determine the tag and word in cache memory using Fully Associative mapping. [2 Marks]

Tag	Word
11 bits	2 bits

Figure 7

13 bit

i) $71813 = 18$

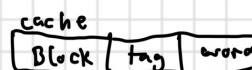
$2^8 \times 16 = 4194304$

ii) total cache word = 2^8
 $\rightarrow$ look at block = 256

iii) tag + (# cache word x size)
 $= 7 + (256 \times 16)$
 $= 4103$

c) 167B: 1 0110 0111 1011 16 bit

tag: 1000 0111 10
word: 11



(d) Suppose a single cache fronting a main memory. The access time of the main memory and the cache is 250 ns and 15 ns respectively. Calculate the Effective Access Time (EAT) given the hit rate is 95%. [2 Marks]

$$d) (H \times \text{access}_c) + ((1 - H) \times \text{access}_{mm})$$

$$(0.95 \times 15) + ((1 - 0.95) \times 250)$$

$$= 26.75 \text{ ns}$$

Effective Access Time (EAT)

The performance of a hierarchical memory is measured by its EAT, or the average time per access.

EAT is a weighted average that uses the hit ratio and the relative access times of the successive levels of the hierarchy.

$$EAT = (H \times \text{Access}_c) + ((1 - H) \times \text{Access}_{mm})$$

H: Cache hit rate
Access_c: Access time for cache
Access_{mm}: Access time for main memory

This formula can be extended to apply to three or more levels of memory.

Example 23

Suppose the cache access time is 15ns, main memory access time is 250ns, and the cache hit rate is 95%.

The average time for the processor to access an item in this two-level memory would then be:

$$EAT = (H \times \text{Access}_c) + ((1 - H) \times \text{Access}_{mm})$$

$$= (0.95 \times 15) + ((1 - 0.95) \times 250)$$

$$= 26.75 \text{ ns}$$

- (d) Suppose a single cache fronting a main memory. The access time of the main memory and the cache is 250 ns and 15 ns respectively. Calculate the Effective Access Time (EAT) given the hit rate is 95%. [2 Marks]

QUESTION 6 [14 Marks]

- (a) List the advantages of Direct Memory Access (DMA) as opposed to Interrupt-driven I/O? [4 Marks]

(b) **Figure 8**

which is
period of

- (a) List the advantages of Direct Memory Access (DMA) as opposed to Interrupt-driven I/O? [4 Marks]

- (b) **Figure 8** illustrates the bus timing diagram for writing data to the disk drive operations which is executed by DMA and controller. Determine the process for the following period of clocks. [3 Marks]

(i) $t_0 - t_2$

(ii) $t_3 - t_7$

(iii) t_8

(i) $t_0 -$

(ii) $t_3 -$

(iii) t_8

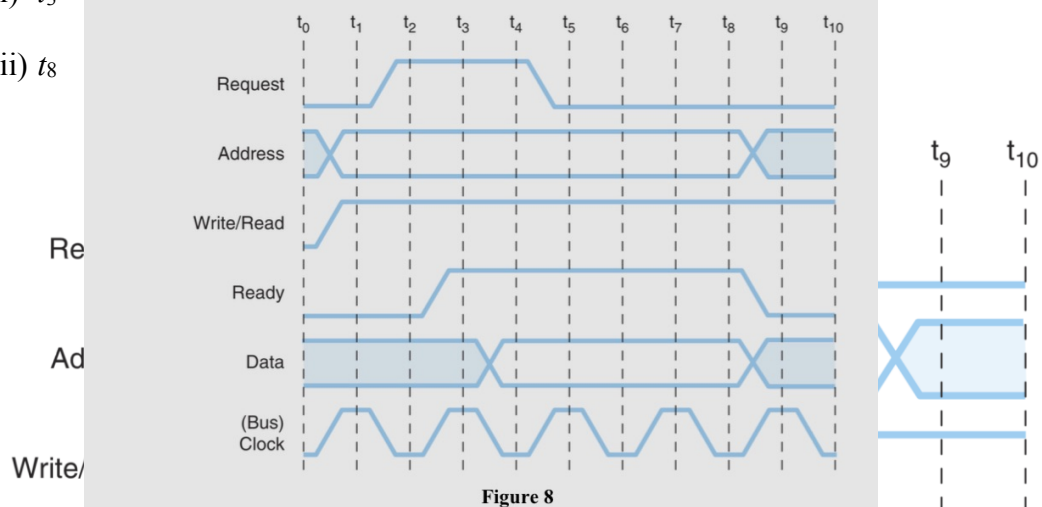


Figure 8

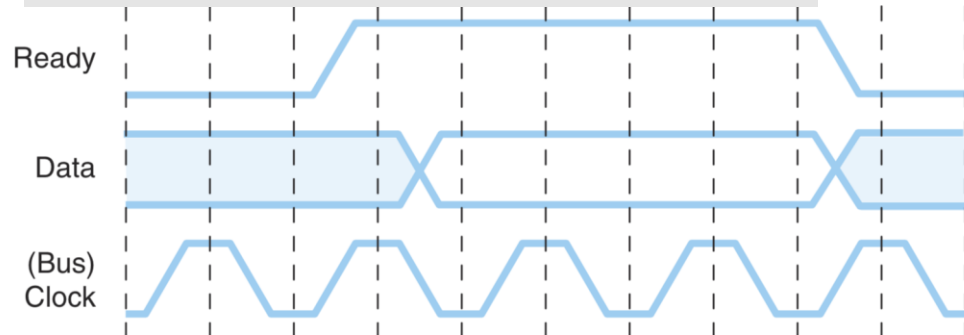


Figure 8

(a) List the advantages of Direct Memory Access (DMA) as opposed to Interrupt-driven I/O? [4 Marks]

(b) Figure 8 illustrates the bus timing diagram for writing data to the disk drive operations which is executed by DMA and controller. Determine the process for the following period of clocks. [3 Marks]

- a). Reduces CPU workload
- Faster data transfer
 - Allows CPU to perform other task
 - lower interrupt overhead

b) i) $t_0 - t_2$ bus is needed for writing
 t_1 - Address - indicates where bytes will

(i) $t_0 - t_2$

(ii) $t_3 - t_7$

(iii) t_8

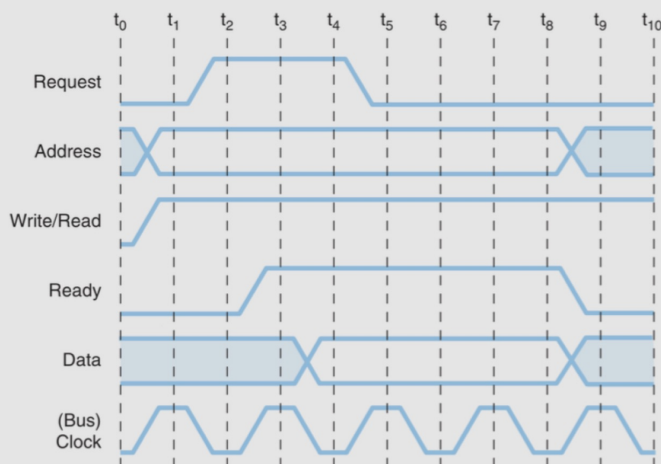


Figure 8

XOR even - 0
 odd - 1

(c) Consider the four data disks and one parity disk are used in RAID Level 3 as shown in Figure 9. Determine the bits for data regeneration labeled as (i) to (iv). [2 Marks]

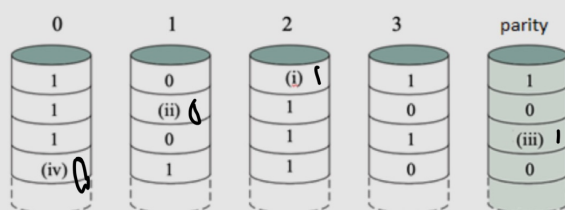


Figure 9

- c) i) 1
 ii) 0
 iii) 1
 iv) 0

(d) Consider the five data disks are used in RAID Level 5 (Block-Level Distributed Parity) as shown in Figure 10. It has four blocks of data and one parity block.

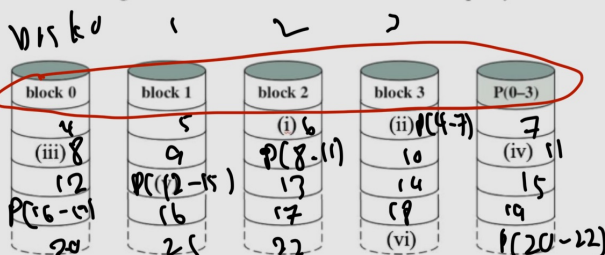


Figure 10

i) RAID (v) - All parity block stored in 1 disk which can become bottleneck during writes.

iv) 5 - parity block is rotated among all disks, eliminating the bottleneck in RAID 4

- ii) i) 6
 ii) P(4-7)
 iii) 8
 iv) 11
 v) P(12-15)
 vi) empty

i) Briefly describe the implementation of the parity strips of this RAID Level 5 compared to RAID Level 4. (Remark: Identify the related scheme used). [2 Marks]

block 0 - block 22

ii) Consider that 23 blocks are required to be added in the disks. Determine the content of data blocks and parity block labeled as (i) to (vi). [3 Marks]

- (c) Consider the four data disks and one parity disk are used in RAID Level 3 as shown in **Figure 9**. Determine the bits for data regeneration labeled as (i) to (iv). [2 Marks]

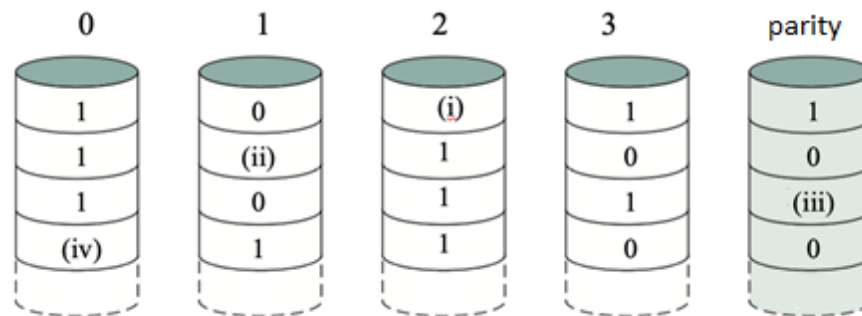


Figure 9

- (d) Consider the five data disks are used in RAID Level 5 (Block-Level Distributed Parity) as shown in **Figure 10**. It has four blocks of data and one parity block.

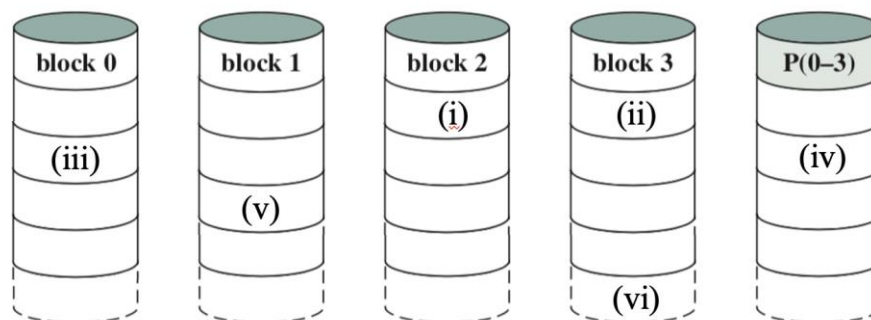


Figure 10

- Briefly describe the implementation of the parity strips of this RAID Level 5 compared to RAID Level 4. (Remark: Identify the related scheme used). [2 Marks]
- Consider that 23 blocks are required to be added in the disks. Determine the content of data blocks and parity block labeled as (i) to (vi). [3 Marks]

QUESTION 7 [15 Marks]

- (a) **Table 9** shows the information on program A that needs to be executed by computer X. Given the clock cycle time for computer X is 200 ps and the total instructions of program A are 1 million.

Table 9

Instruction type	CPI	Instruction mix (%)	Clock cycles
Arithmetic	3	30	(i)
Load/Store	2	50	(ii)
Branch	4	20	(iii)

- Calculate the clock cycles for each Instruction type and write the answers for (i), (ii) and (iii) as in **Table 9**. [2 Marks]
 - Calculate the average CPI and the CPU execution time. [3 Marks]
 - Calculate the CPU Clock rate and MIPS rate. [2 Marks]
- (b) **Table 10** shows the Execution frequency and time for 3 different programs run by CPU A, B and C.

Table 10

Program	Execution frequency (%)	Execution time (ns)		
		CPU A	CPU B	CPU C
P	50	1000	900	800
Q	30	650	300	200
R	20	200	100	50

- Calculate the Weighted Average for CPU A, B and C. [3 Marks]
- Identify the fastest CPU and how many percentage (%) it is faster than the slowest CPU. [2 Marks]
- Assume that CPU C is normalized to CPU A, calculate the Geometric Mean for CPU C. [3 Marks]

--- End of Questions ---

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- i) Calculate the clock cycles for each Instruction type and write the answers for (i), (ii) and (iii) as in **Table 9**. [2 Marks]
- ii) Calculate the average CPI and the CPU execution time. [3 Marks]
- iii) Calculate the CPU Clock rate and MIPS rate. [2 Marks]

i) $3 \times 0.3 = 0.9$

ii) $2 \times 0.5 = 1$

iii) $4 \times 0.2 = 0.8$

i) $0.9 + 1 + 0.8 = 2.7 \rightarrow$ average CPI

CPU execution time = $I_c \times CPI \times CCT$
 $= 1000000 \times 2.7 \times 200 \times 10^{-12}$
 $= 5.4 \times 10^{-4}$
maybe same as CPU time

iii) clock cycle time = $\frac{1}{\text{CPU clock rate}} = \frac{1}{200 \times 10^6} = 5 \text{ GHz}$

MIPS rate = $\frac{f}{CPI \times 10^6} = \frac{5 \text{ GHz}}{2.7 \times 10^6}$
 $= 1851.85 \text{ MIPS}$

(b) **Table 10** shows the Execution frequency and time for 3 different programs run by CPU A, B and C.

Table 10

Program	Execution frequency (%)	Execution time (ns)		
		CPU A	CPU B	CPU C
P	50	1000	900	800
Q	30	650	300	200
R	20	200	100	50

- i) Calculate the Weighted Average for CPU A, B and C. [3 Marks]
- ii) Identify the fastest CPU and how many percentage (%) it is faster than the slowest CPU. [2 Marks]
- iii) Assume that CPU C is normalized to CPU A, calculate the Geometric Mean for CPU C. [3 Marks]

i) $(1000 \times 0.5) + (650 \times 0.3) + (200 \times 0.2)$

A = 735
B = 566
C = 470

ii) $\left(\frac{735}{470} - 1 \right) \times 100 = 56.38\%$

C 56.38% faster than A

iii) $\sqrt[3]{1.25 \times 3.25 \times 4.00}$
 $= 2.53$

C normalized to CPU A is approximately 2.53

A C
1 1.25
3 3.25
4 4.00

Appendix A

(Microinstruction fields)

F1	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC + DR$	ADD
010	$AC \leftarrow 0$	CLRAC
011	$AC \leftarrow AC + 1$	INCAC
100	$AC \leftarrow DR$	DRTAC
101	$AR \leftarrow DR(0-10)$	DRTAR
110	$AR \leftarrow PC$	PCTAR
111	$M[AR] \leftarrow DR$	WRITE

F2	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC - DR$	SUB
010	$AC \leftarrow AC \vee DR$	OR
011	$AC \leftarrow AC \wedge DR$	AND
100	$DR \leftarrow M[AR]$	READ
101	$DR \leftarrow AC$	ACTDR
110	$DR \leftarrow DR + 1$	INCDR
111	$DR(0-10) \leftarrow PC$	PCTDR

F3	Microoperation	Symbol
000	None	NOP
001	$AC \leftarrow AC \oplus DR$	XOR
010	$AC \leftarrow AC'$	COM
011	$AC \leftarrow \text{shl } AC$	SHL
100	$AC \leftarrow \text{shr } AC$	SHR
101	$PC \leftarrow PC + 1$	INCP
110	$PC \leftarrow AR$	ARTPC
111	Reserved	

CD	Condition	Symbol	Comments
00	Always = 1	U	Unconditional branch
01	$DR(15)$	I	Indirect address bit
10	$AC(15)$	S	Sign bit of AC
11	$AC = 0$	Z	Zero value in AC

BR	Symbol	Function
00	JMP	$CAR \leftarrow AD$ if condition = 1 $CAR \leftarrow CAR + 1$ if condition = 0
01	CALL	$CAR \leftarrow AD, SBR \leftarrow CAR + 1$ if condition = 1 $CAR \leftarrow CAR + 1$ if condition = 0
10	RET	$CAR \leftarrow SBR$ (Return from subroutine)
11	MAP	$CAR(2-5) \leftarrow DR(11-14), CAR(0,1,6) \leftarrow 0$